

2 INTRODUCTION

2.1 Summary

This chapter describes the special terminology used in this manual, installation specifications, configurations and operation outlines.

2.2 Terminology

Some of the special terms used in this manual are explained below.

Term	Definition
Product	The article which is being checked.
Packer	The upstream equipment feeds product to the seal checker.
Empty pack	A sealed pack without product filled.
Chained pack	More than 2 packs are connected together due to no cut with a packer.
Broken bag	A bag broken product by entanglement into packer

2.3 Specifications

Specifications are shown in the table below.

NOTE

- The following specifications are for standard model and it may differ from that for the seal checker delivered to you.
- The air supply setting is applicable for air jets used in the rejection of NG products.

Item	Specification	
Model	TSC-RS	
Min. thickness indication	0.01mm	
Product	Pillow package	
Product size	Width	50 - 330mm
	Length	100 - 650mm
	Thickness	15 - 120mm
Conveyor speed set range	J-belt conveyor	Ten-key setting 15-90m/min (at interval of 1m)
	Infeed conveyor	Ten-key setting 15-70m/min (at interval of 1m)
	Brush Conditioner	Ten-key setting 15-70m/min (at interval of 1m)
	Seal check conveyor	Ten-key setting 15-70m/min (at interval of 1m)
	Reject conveyor	Ten-key setting 15-90m/min (at interval of 1m)
Main body size	Width 800mm×1500mm×Length 2800mm	
Weight	Approx. 220kg	
Power supply	Voltage/ Frequency	Single phase: AC200V, 50/60Hz Single phase: AC208V, 50/60Hz Single phase: AC220V, 50/60Hz Single phase: AC230V, 50/60Hz Single phase: AC240V, 50/60Hz
	Allowable voltage variation	Single phase AC200V: AC180-220V Single phase AC208V: AC187-229V Single phase AC220V: AC198-242V Single phase AC230V: AC207-253V Single phase AC240V: AC216-264V
	Allowable frequency variation	Within±5%
	Power installation capacity	0.5kW 9.8A(AC208V)
Air source	Air pressure	0.4-1.0MPa
	Air consumption	6.4 NI/min. (ANR) When reject discharge

Item	Specification	
Environment	Ambient temperature	0-40°C(no freezing)
	Humidity	30-85%RH(no dew condensation)
	Atmosphere	Indoor(No direct sunlight permitted), free of corrosive gas, inflammable gas, oil mist and dust

2.4 Configuration of the System

Name of each part and the function of the system are described below:

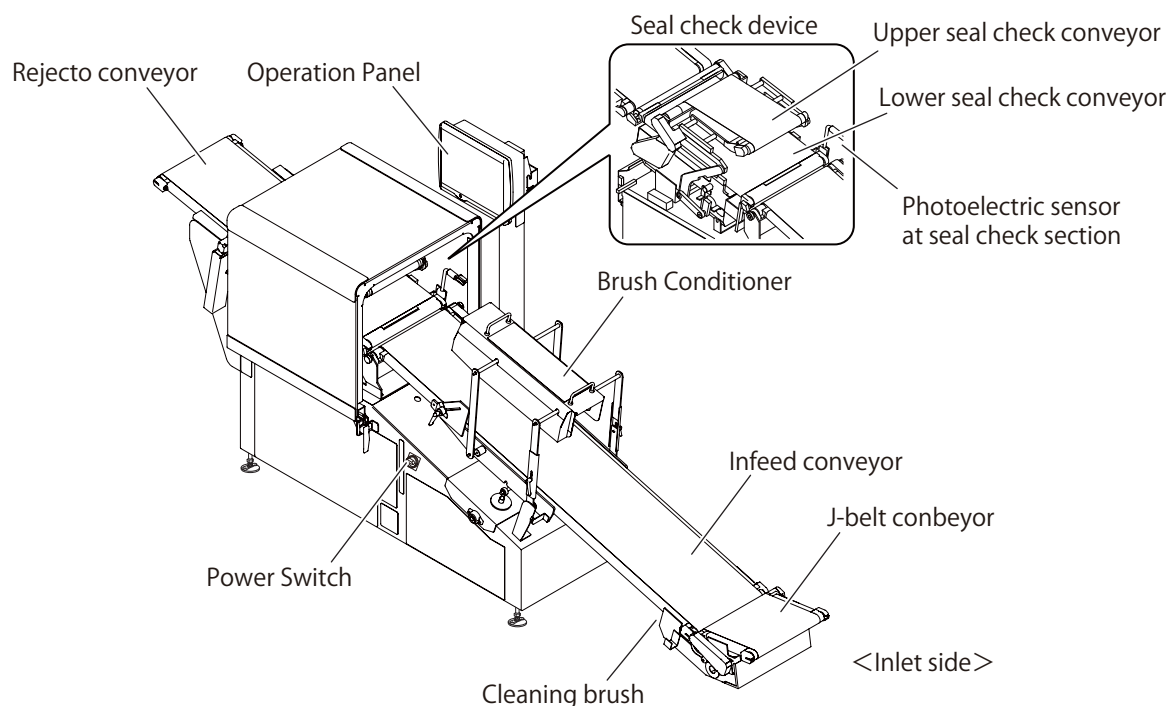


Fig.2-1 Appearance of the system

Table 2-1 Each Part of the System

Name	Function and Description
Power switch	Power supply to the system is started or shut off.
Operation panel	System operation or setting of operating conditions is performed from the touch panel. Operating status and setting status are also displayed.
Printer(option)	Total data of measurements are printed.
J-belt conveyor	Products discharged from packer are supplied onto the infeed conveyor. Conveyor is driven by motor and forcibly carries products to prevent retention.
Photoelectric sensor at pre-rejector	It detects that a product is carried onto the infeed conveyor. It also detects a broken bag of product caused by entanglement into the packer.
Infeed conveyor	Products carried by the J-belt conveyor are transferred onto the lower seal check conveyor. Brush conditioner is provided above the discharge side.
Pre-rejector air jet(option)	A broken bagged product is removed by compressed air.
Brush Conditioner	Conditioning conveyor which is installed on the discharge side of infeed conveyor. The form of product carried into the seal check device is preliminarily adjusted by the dead weight of conditioning conveyor. Brush Conditioner flattens the bag thickness by pushing the bag with rotating roll brush.
Photoelectric sensor at seal check section(option)	This detects that a product has been carried into the seal check device.
Cleaning brush (option)	This removes dirt and dust on the belt surface of belt conveyor.

Table 2-1 Each Part of the System (Continued)

Name	Function and Description
Lower seal check conveyor	This carries a product to be seal check.
Seal check device	This measures the product bag thickness to be seal check.
Upper seal check conveyor	Load is applied to the passing product to measure the change of product thickness to obtain the determination data for the quality of seal.
Reject conveyor	Product accepted/rejected at the seal check section is carried to the post process.
Rejector air jet	Accepted/rejected product after completion of seal check is distributed by compressed air.

2.5 Operation Outline

This section describes a series of product flow from feeding into this system through measurement/discharge.

NOTE

- The figure shown below is of a equipment with the air jet type rejector device.

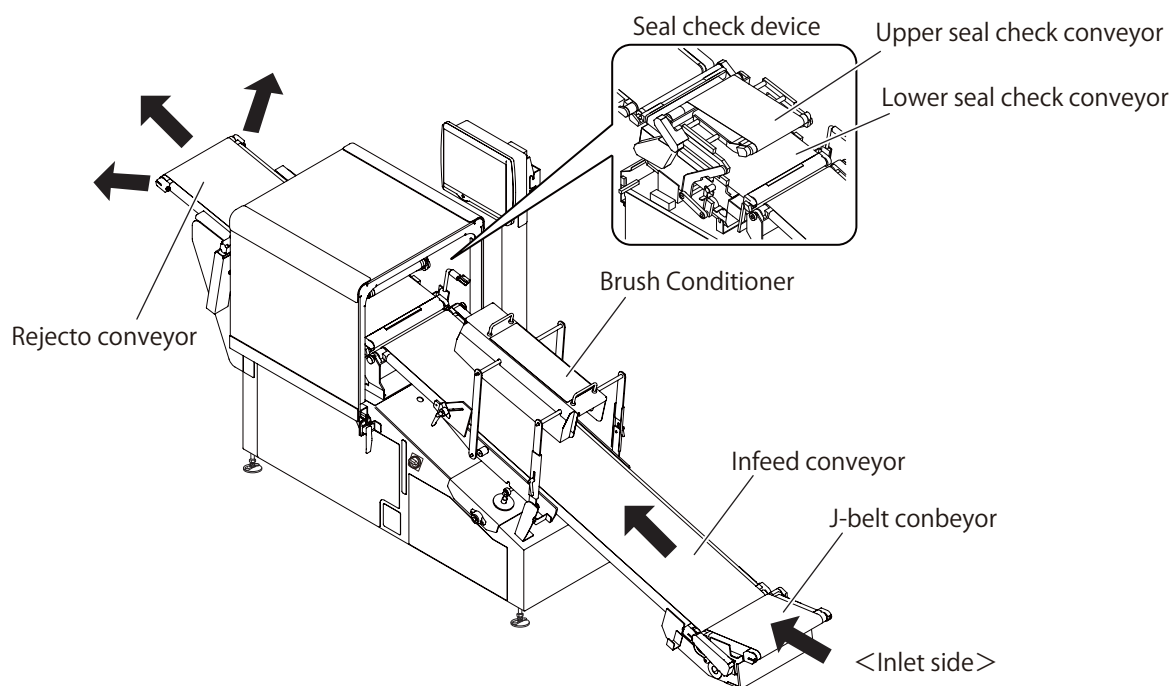


Fig.2-2 Operation Schematic Diagram

1. Product is fed onto the J-belt conveyor. (In the case of a broken bag, it is removed by the pre-rejector(option).)
2. The infeed conveyor carries product.
3. Product is carried onto the lower seal check conveyor.
4. The photoelectric sensor at the seal check section detects that the product is carried into the seal check device.
5. The upper seal check conveyor measures the product thickness.
6. Measurements are determined acceptable or rejectable based on the set value from the operation panel.
7. Based on the judgement of product to be acceptable or rejectable, it is distributed in the direction set by the operation panel.